==========

INTERNSHIP PROJECT PROPOSAL

TRACK: MACHINE LEARNING & DATA SCIENCE

==========

APPLICANT DETAILS

- **Intern Name:** **Mehmood Hassan**
- **Intern ID:** **JWS-2026-ML-006**
- **Institute:** **Govt. College of Technology Printing and Graphic Arts, Lahore**
- **Submitted To:** **Jamwah Studios**
- **Date:** **September 10, 2026**

==========

1. PROJECT OVERVIEW & DEFINITION

- ***Project Title:*** ***Enterprise Customer Churn Prediction System***

- **Domain Area:**
  Predictive Analytics & Supervised Classification (Customer Retention)

- **Core Objective:**

The primary objective of this project is to architect an end-to-end Machine Learning classification pipeline that evaluates customer demographics, account tenure, usage patterns, and support history. The system aims to accurately predict churn risk, stratify customers into risk tiers, identify key churn drivers via Explainable AI (XAI), and serve inference through a Dockerized FastAPI service.

---

2. DATASET SPECIFICATIONS & PREPROCESSING PIPELINE

- ***Data Source:***
  Kaggle (Customer Churn Dataset - Shahid Azeem).

- ***Dataset Features & Architecture:***
  - Key Attributes: Age, tenure (months), monthly charges, total charges, complaints count, usage level, contract type, and payment methods.
  - Record Volume: ~64,000+ comprehensive customer records across multi-part datasets.
  - Target Output: Customer Churn Status (1 = Churned vs. 0 = Retained/Stayed).

- ***Data Preprocessing Strategy:***
  - Imputation: Handling missing or null entries via dropna, median and mode imputation.
  - Feature Encoding: Transforming categorical variables using One-Hot and Label Encoding.

- Normalization: Applying StandardScaler to normalize numerical feature values.
- Dimension Reduction: Feature selection using correlation matrix heatmaps and feature importance ranking.

---

3. METHODOLOGY & TECHNICAL INFRASTRUCTURE

• Model Architecture (Algorithms to Evaluate):

- Baseline Classifier: Logistic Regression
- Tree-Based Models: Decision Tree Classifier, Random Forest Classifier
- Advanced Ensembles: XGBoost / Gradient Boosting Classifiers

• Development Stack & Tooling:

- Programming Engine: Python 3.14
- Data Processing: Pandas, NumPy
- Visual Analytics: Matplotlib, Seaborn
- Machine Learning: Scikit-Learn, XGBoost
- API & Containerization: FastAPI, Uvicorn, Docker
- IDE / Environment: VS Code & Jupyter Notebooks (Conda Environment)

• Performance Evaluation Metrics:

- Classification Accuracy Score
- Precision, Recall, and F1-Score (optimized for capturing high-risk churners)
- Confusion Matrix Analysis & ROC-AUC Curves

==========